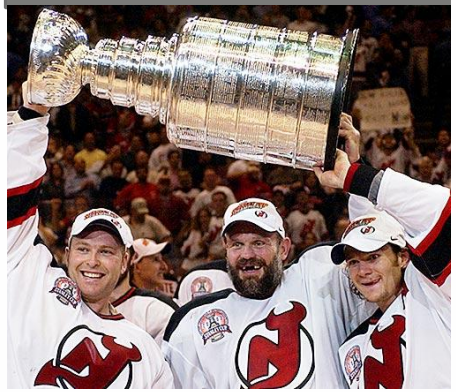
By Olivia Snyder

Analysis of the 2002-03 NHL Season



Final Project for STAT 1341

Summary

30 Teams, 6 Divisions

Atlantic

- New Jersey Devils
- Philadelphia Flyers
- New York Islanders
- New York Rangers
- Pittsburgh Penguins

Northeast

- Ottawa Senators
- Toronto Maple Leafs
- Boston Bruins
- Montreal Canadiens
- Buffalo Sabres

Southeast

- Tampa Bay Lightning
- Washington Capitals
- Atlanta Thrashers
- Florida Panthers
- Carolina Hurricanes

Central

- Detroit Red Wings
- St. Louis Blues
- Chicago Blackhawks
- Nashville Predators
- Columbus Blue Jackets

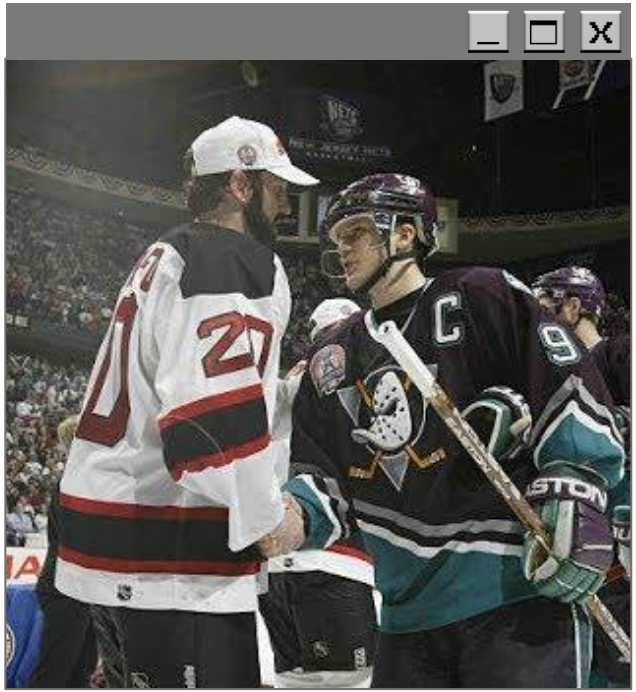
Northwest

- Colorado Avalanche
- Vancouver Canucks
- Minnesota Wild
- Edmonton Oilers
- Calgary Flames

Pacific

- Dallas Stars
- Anaheim Ducks
- Los Angeles Kings
- Phoenix Coyotes
- San Jose Sharks

Summary

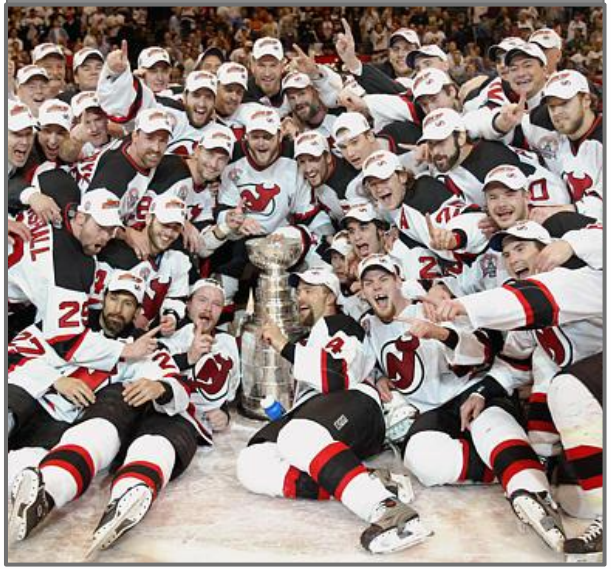


2002-2003 Season Recap: Anaheim Ducks

The 7th-seeded Anaheim Ducks made an unexpected appearance in the Stanley Cup final. They defeated the 2nd-seeded Detroit Red Wings in the Quarterfinals, the top-seeded Dallas Stars in the semifinals, and the sixth-seeded Minnesota Wild in the conference finals. Although the Ducks’ season ended with a loss in the finals to the New Jersey Devils, Jean-Sébastien Giguère's playoff performance won him the Conn Smythe Trophy. Their Cinderella Run makes this season’s predictive outcomes particularly interesting.

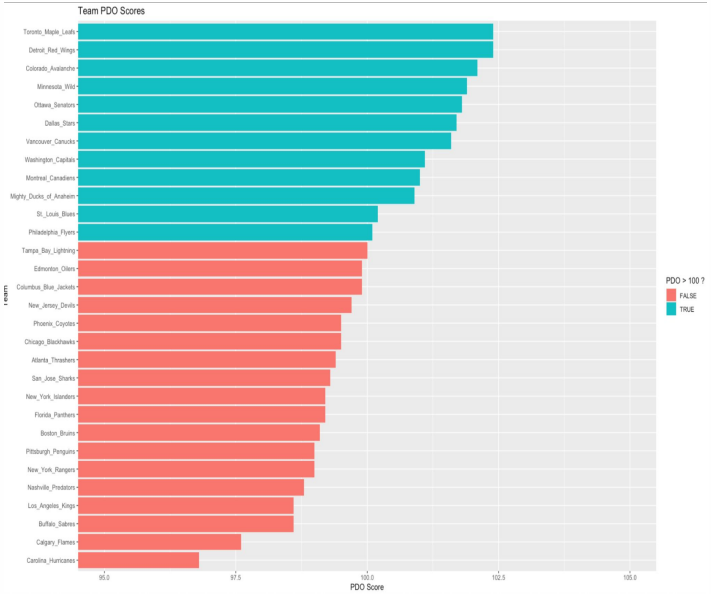
Season Recap: Notable Events

- New Jersey used a defensive “trap” strategy to minimize their goals against. They finished the season as Stanley Cup winners with the second lowest goals against in the league.
- The ‘02-’03 season ended with large scoring disparities between NHL teams; The Ottawa Senators finished with a total of 113 points, while the Carolina Hurricanes ended with only 61 points.



Relevant Statistics: PDO

Mean: 100.01 SD: 1.415735
Min: 96.8 Max:102.4



PDO

PDO is a metric combining a team's shooting percentage and save percentage, serving as a key indicator of "puck luck". A score of ~100 is the average; scores above 100 suggest good luck, and scores below 100 suggest bad luck. The league average PDO was 100.01, confirming that the league did not experience any long-term luck deviation. The small standard deviation (1.416735) suggests that most teams had a relatively normal distribution of luck across the season, with most teams very close to the league mean. During the 2002-2003 season, the Carolina Hurricanes had the lowest PDO (96.8), indicating they experienced the worst luck in the league. The Toronto Maple Leafs had the best luck with a PDO of 102.4.

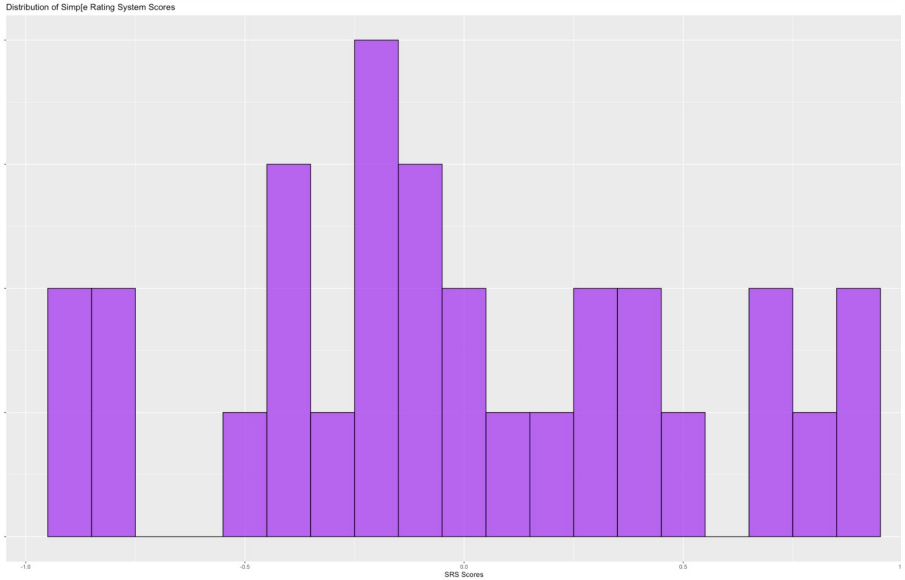
Relevant Statistics: Simple Rating System

The Simple Rating System (SRS) measures a team's strength by its average Goal Differential, adjusted for the strength of its opponent. The league mean for SRS was -7.684283e^-18, verifying that all ratings are calculated relative to the league average of 0. SRS has an average standard deviation of 0.5278584, which suggests the difference between an average and above-average team is relatively small. Top and bottom-tier teams are clearly separated from the middle group, showing three distinct groups based on SRS scores. The Dallas Stars had the highest SRS (0.94), indicating they are a part of the elite group that dominated the score differential. Comparatively, the Carolina Hurricanes had the lowest SRS, putting them at an expected -0.9 goals lost on average against an opponent.

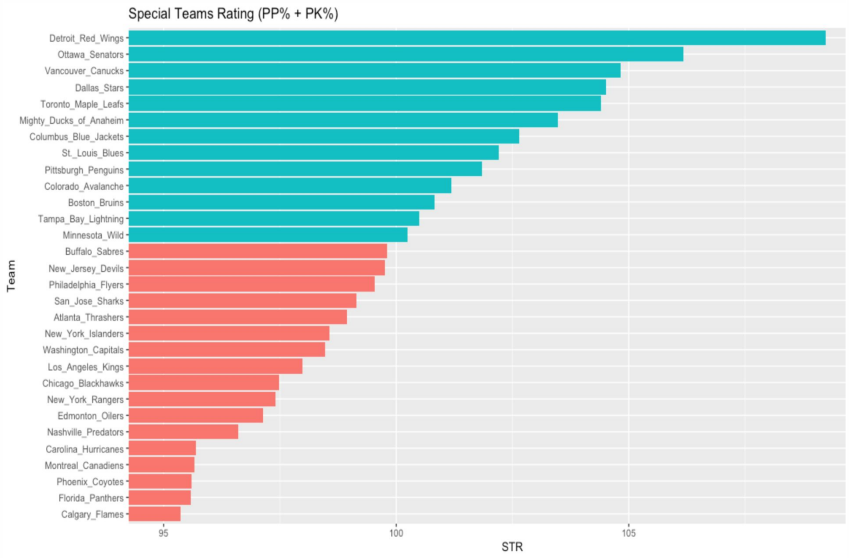
Mean: -7.684283e^-18 **SD: 0.5278584**

Min: -0.9 **Max: 0.94**

Simple Rating System (SRS)



Relevant Statistics: Special Teams Rating



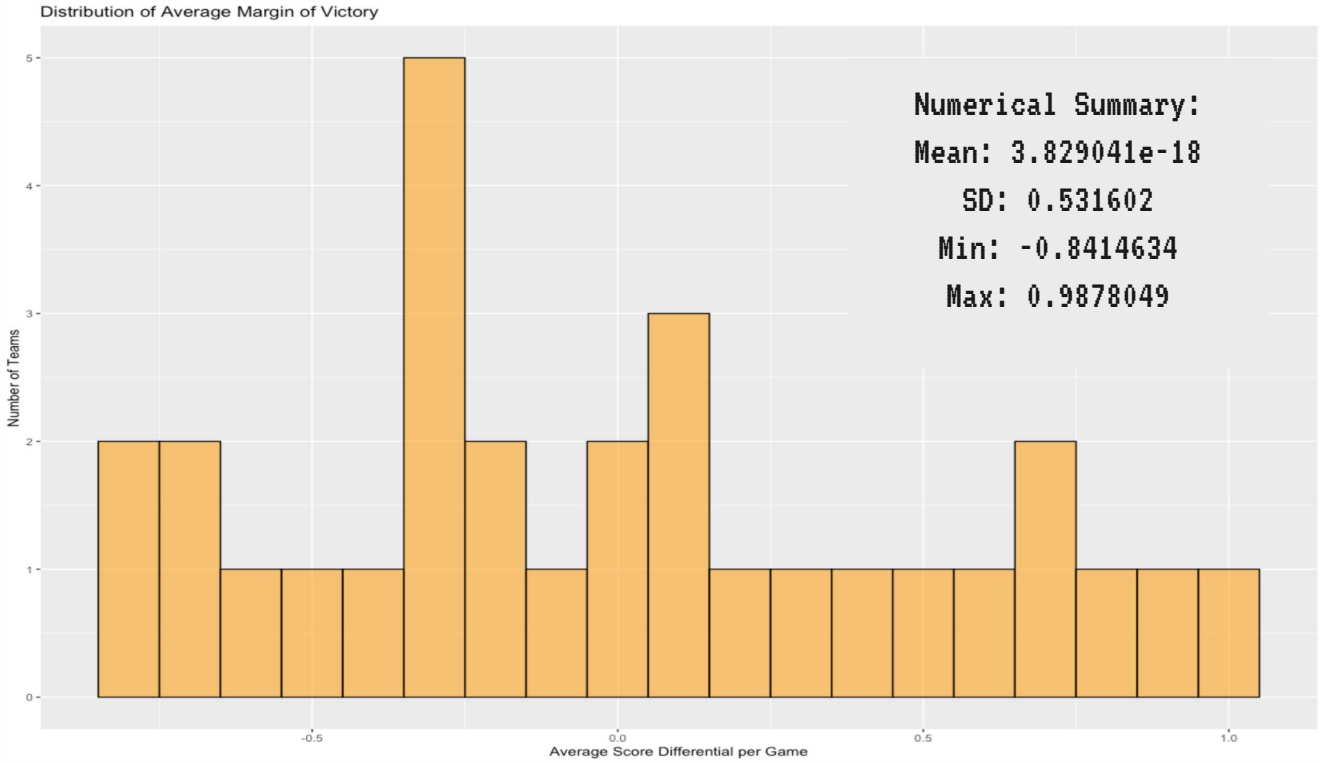
Mean: 100.01 **SD: 1.415735**

Min: 96.8 **Max:102.4**

Special Teams Rating (STR)

The Special Teams Rating is derived from power play percentage plus penalty kill percentage. STR is used as a measure of a team's efficiency in situations where one team has an advantage. Similar to PDO, the STR league mean (100.0267) being near 100 suggests that the average team was balanced in their performance between power plays and penalty kills. Whatever advantages the team makes on a power play are cancelled out by the disadvantages of the penalty kill with an average STR. The standard deviation (3.508958) shows that special teams efficiency varies more than PDO, suggesting some teams at the top have a significant advantage over bottom teams. The Calgary Flames had the lowest rating (95.37) compared to the Detroit Red Wings at the top (109.23), indicating the Red Wings' elite status.

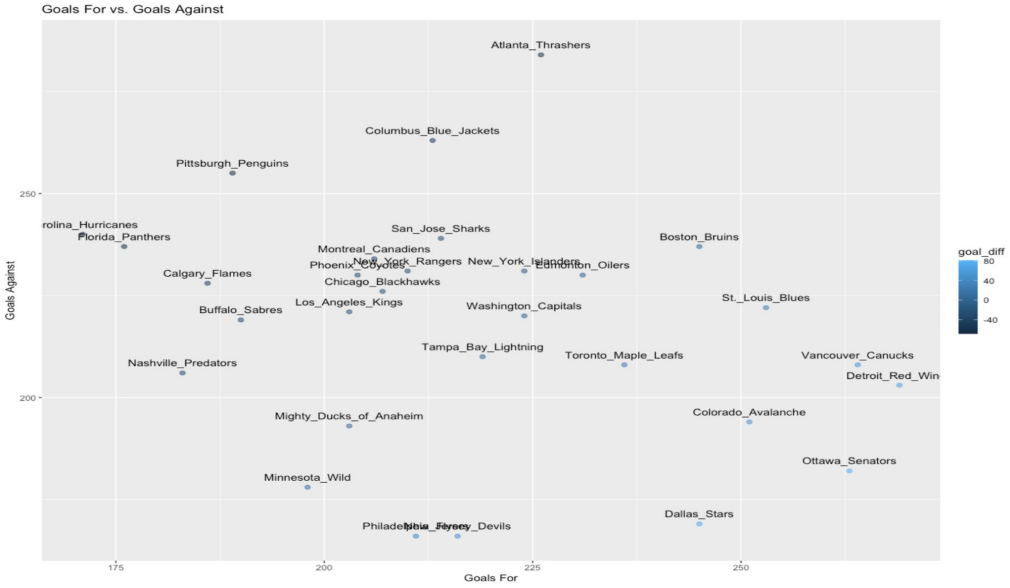
Scoring Distribution



Scoring Distribution

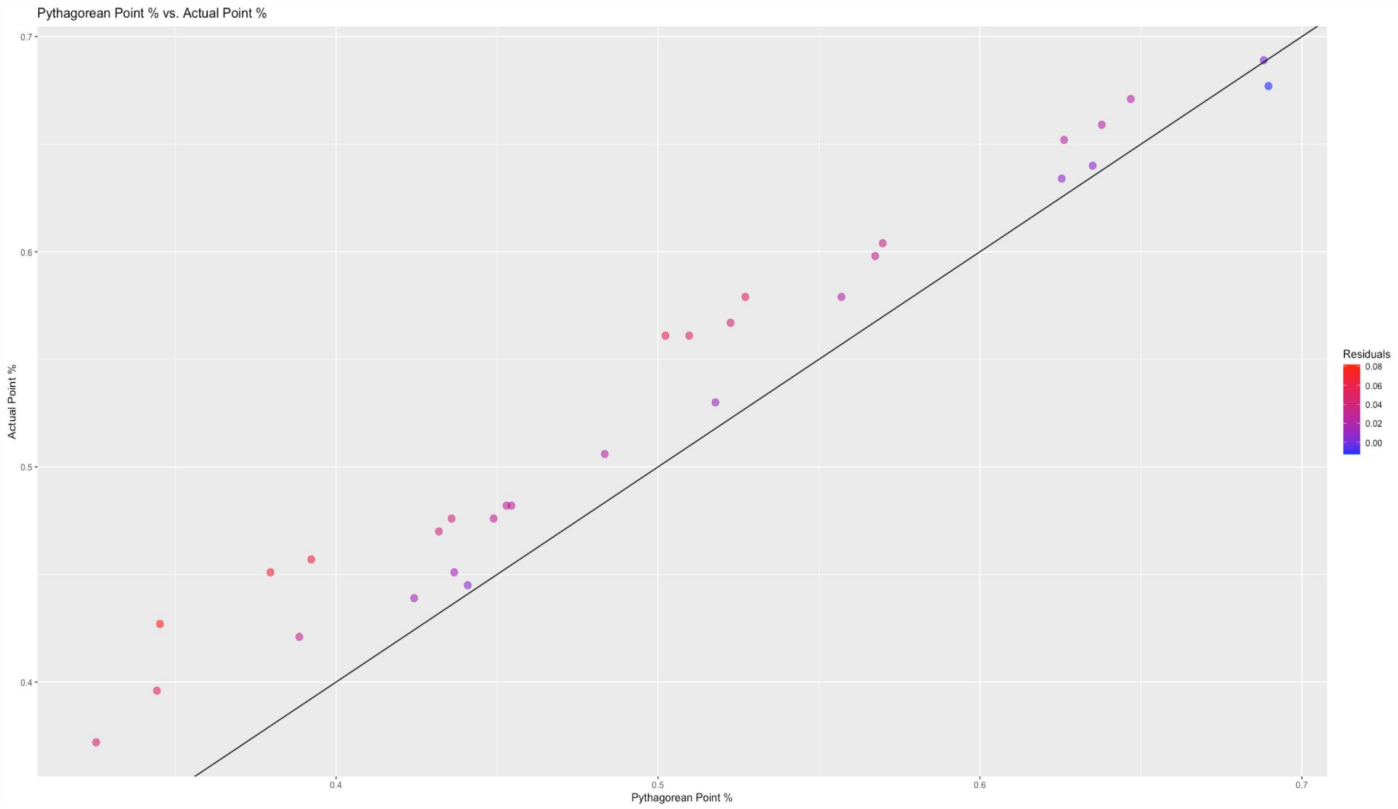
Goals For vs. Goals Against

Correlation: -0.2682778



The distribution of the Margin of Victory (MoV) reveals key components of the league's competitive balance. The mean MoV of 3.829041e-18 confirms that goals are being measured as a zero-sum statistic; a goal for is always a goal against. The standard deviation of 0.531602 indicates that the typical MoV between a team and the league average is roughly half a goal per game, suggesting a wide variation in scoring ability with a cluster around the middle. The histogram shows this clustering towards the middle, with the elite and underachieving teams creating peaks at the tails. A weak, negative correlation between Goals For (GF) and Goals Against (GA) of -0.2682778 implies that a team excelling in one area does not guarantee that they will excel in the other. The smallest goal differential belonged to the Carolina Hurricanes at -69 goals and a Margin of Victory of -0.84146341. The Ottawa Senators confirmed their elite status with a goal differential of 81 and an MoV of 0.98780488, outscoring the Hurricanes by nearly two goals per game.

Pythagorean Point Percentage



Numerical Summary

Overachiever: Florida Panthers (0.0817057)
Underachiever: Dallas Stars (-0.01263703)
Residual mean: 0.03147056
Residual Standard Error: 0.02190007
Correlation: 0.9830765

The relationship between Pythagorean point percentage and actual point percentage for this season is extremely strong, as evidenced by the correlation of 0.9830765. Most data points fall relatively close to the reference y = x line, reflecting the small residual standard error of 0.02190007. The residual mean of 0.03147056 reveals a league-wide overachievement relative to goal differential. However, specific teams showed significant deviation. The Florida Panthers were the extreme overachiever with a residual of 0.0817057, meaning they win 8.17% more games than the scoring margins suggested. Conversely, the Dallas Stars were the biggest underachiever in the league with a residual of -0.01263703. Even with their dominant goal differential, their number of wins was lower than expected, possibly due to a worse performance in close games.

Scoreline Probability Distribution

A Poisson Scoring model is appropriate for low-scoring sports, such as hockey. The scoreline probability distribution to the right lists all possible final score outcomes for a regular season game and their corresponding likelihoods. The table lists a 2-1 Devils victory as the most probable outcome for a regular season game, with a likelihood of 8.05%.

Anaheim Ducks

New Jersey Devils

	0	1	2	3	4	5	6	7	8
0	1.82	3.23	2.86	1.69	0.75	0.27	0.08	0.02	0.00
1	4.07	7.21	6.39	3.78	1.67	0.59	0.18	0.04	0.01
2	4.54	8.05	7.13	4.21	1.87	0.66	0.20	0.05	0.01
3	3.38	5.99	5.31	3.14	1.39	0.49	0.15	0.04	0.01
4	1.89	3.34	2.96	1.75	0.78	0.27	0.08	0.02	0.00
5	0.84	1.49	1.32	0.78	0.35	0.12	0.04	0.01	0.00
6	0.31	0.56	0.49	0.29	0.13	0.05	0.01	0.00	0.00
7	0.010	0.18	0.16	0.09	0.04	0.01	0.00	0.00	0.00
8	0.03	0.05	0.04	0.03	0.01	0.00	0.00	0.00	0.00

Poisson Rating

The Poisson model captures a team's offensive and defensive abilities. Higher positive coefficients represent teams that are better at both, while a negative coefficient indicates a worse performance. Based on the coefficients from the Poisson model, the Detroit Red Wings were arguably the strongest team; they had an offensive coefficient of 0.195579 and a defensive coefficient of 0.363192. On the contrary, the Carolina Hurricanes had the most negative coefficient (-0.270145), which aligns with their low SRS and minimum score differential. Because all defensive coefficients are positive, the Winnipeg Jets fall behind as the worst defensive team. The Jets were used as the baseline category, implicitly setting their defensive coefficient to 0.0. Any positive defensive coefficient indicates a better performance than the Jets, and any negative defensive coefficient suggests worse.

Poisson Rating

Team	Offensive Coefficient	P-value (Offensive)	Defensive Coefficient	P-value (Defensive)
Anaheim Ducks	-0.09434713	0.341827	0.438635	4.48e-06***
Boston Bruins	0.08069385	0.389468	0.180467	0.042522*
Buffalo Sabres	-0.171516	0.085526	0.298029	0.001112**
Calgary Flames	-0.1700446	0.094231	0.270282	0.002924**
Carolina Hurricanes	-0.270145	0.008467**	0.187981	0.034342*
Chicago Blackhawks	-0.057782	0.557233	0.276220	0.002474**
Colorado Avalanche	0.122636	0.192770	0.435931	5.24e-06***
Columbus Blue Jackets	-0.019694	0.840402	0.107283	0.219774

Team	Offensive Coefficient	P-value (Offensive)	Defensive Coefficient	P-value (Defensive)
Dallas Stars	0.095319	0.313366	0.551551	2.76e-08***
Detroit Red Wings	0.195579	0.034779*	0.363192	0.000109***
Edmonton Oilers	0.042896	0.655553	0.275206	0.002555**
Florida Panthers	-0.249176	0.014692*	0.220310	0.013836*
Los Angeles Kings	-0.087114	0.380130	0.304440	0.000937***
Minnesota Wild	-0.125806	0.209039	0.494784	3.51e-07***
Montreal Canadiens	-0.078818	0.418842	0.226482	0.011655*
Nashville Predators	-0.194854	0.055877	0.389970	3.25e-05***

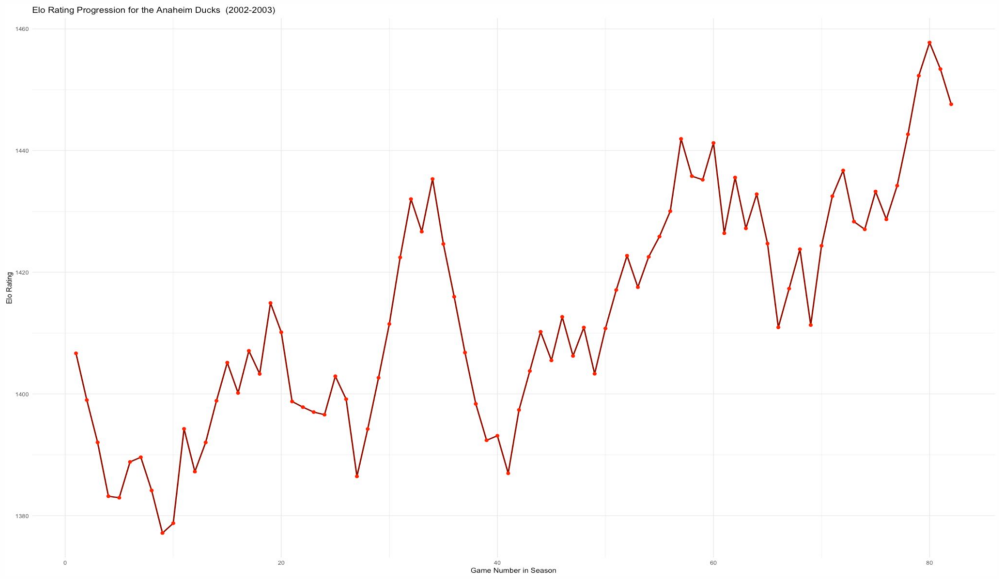
Poisson Rating

Team	Offensive Coefficient	P-value (Offensive)	Defensive Coefficient	P-value (Defensive)
New Jersey Devils	-0.076672	0.429252	0.558783	1.86e-08***
New York Islanders	-0.008591	0.928518	0.203068	0.023066*
New York Rangers	-0.079824	0.412963	0.209967	0.019078*
Ottawa Senators	0.140738	0.127044	0.431375	6.46e-06***
Philadelphia Flyers	-0.091081	0.349525	0.548625	2.95e-08***
Pittsburgh Penguins	-0.163946	0.100843	0.119216	0.172017
San Jose Sharks	-0.006861	0.943818	0.224268	0.012748*

Team	Offensive Coefficient	P-value (Offensive)	Defensive Coefficient	P-value (Defensive)
St. Louis Blues	0.138879	0.139438	0.281287	0.002187**
Tampa Bay Lightning	-0.028695	0.765089	0.305113	0.000939***
Toronto Maple Leafs	0.028675	0.762134	0.317675	0.000572***
Utah Hockey Club	-0.069370	0.482878	0.250184	0.005841**
Vancouver Canucks	0.186338	0.044863*	0.323841	0.000486***
Washington Capitals	-0.022094	0.817828	0.264085	0.003748**
Winnipeg Jets	NA	NA	NA	NA

The Anaheim Ducks' season was characterized by instability, as evidenced by the high variability in their Elo ratings. There are four distinct peaks in their season Elo ratings; two before the middle of the season and two after. There are four subsequent drops in between each peak, indicating the Ducks' pattern of long winning streaks followed by hard losses. The Ducks reached their global maximum Elo (1457.75) at game 80 after a 2-1 win against the Nashville Predators, gaining them an adjustment of 5.46. The largest elo adjustment they received during the entire season was 15.52 after a 4-1 victory over the Boston Bruins (Game 11). Conversely, the largest adjustment the Ducks lost was -14.84 after a 2-6 loss against the New York Rangers (Game 61). The Ducks reached their highest win probability (0.6185) against the Rangers in game 61, which tanked their Elo after the unexpected loss. Despite the extreme mid-season volatility, the Ducks finished their season with an impressive 1447.61 and a qualification for the Stanley Cup playoffs.

Analysis of Anaheim Ducks Elo Ratings



Simulated Win Totals

The simulation analysis clearly identifies teams that under- or over-performed their Elo expectations. The Montreal Canadiens exhibited the greatest underperformance, with their 34 actual wins being 8.315 lower than their simulated wins, creating the widest win disparity and confirming their weak season. On the other hand, the Ottawa Senators were the league’s biggest overachievers, winning 6.637 more games than their expected 49.363 simulated wins. The Utah Hockey Club (Known then as the Phoenix Coyotes) had the most accurate prediction, winning only 0.388 games less than predicted. The Monte Carlo simulation accurately predicted the Dallas Stars, Detroit Red Wings, Tampa Bay Lightning, New Jersey Devils, and Ottawa Senators winning their respective divisions. However, the simulation inaccurately predicted the Vancouver Canucks as the winner of the Northwest division, which was actually won by the Colorado Avalanche.

Monte Carlo Simulation

Team	Actual Wins	Simulated Wins
Anaheim Ducks	44.5	40.287
Boston Bruins	41.5	46.685
Buffalo Sabres	32	34.113
Calgary Flames	35.5	34.340
Carolina Hurricanes	27.5	34.517
Chicago Blackhawks	36.5	43.578
Colorado Avalanche	48.5	44.093
Columbus Blue Jackets	33	35.283

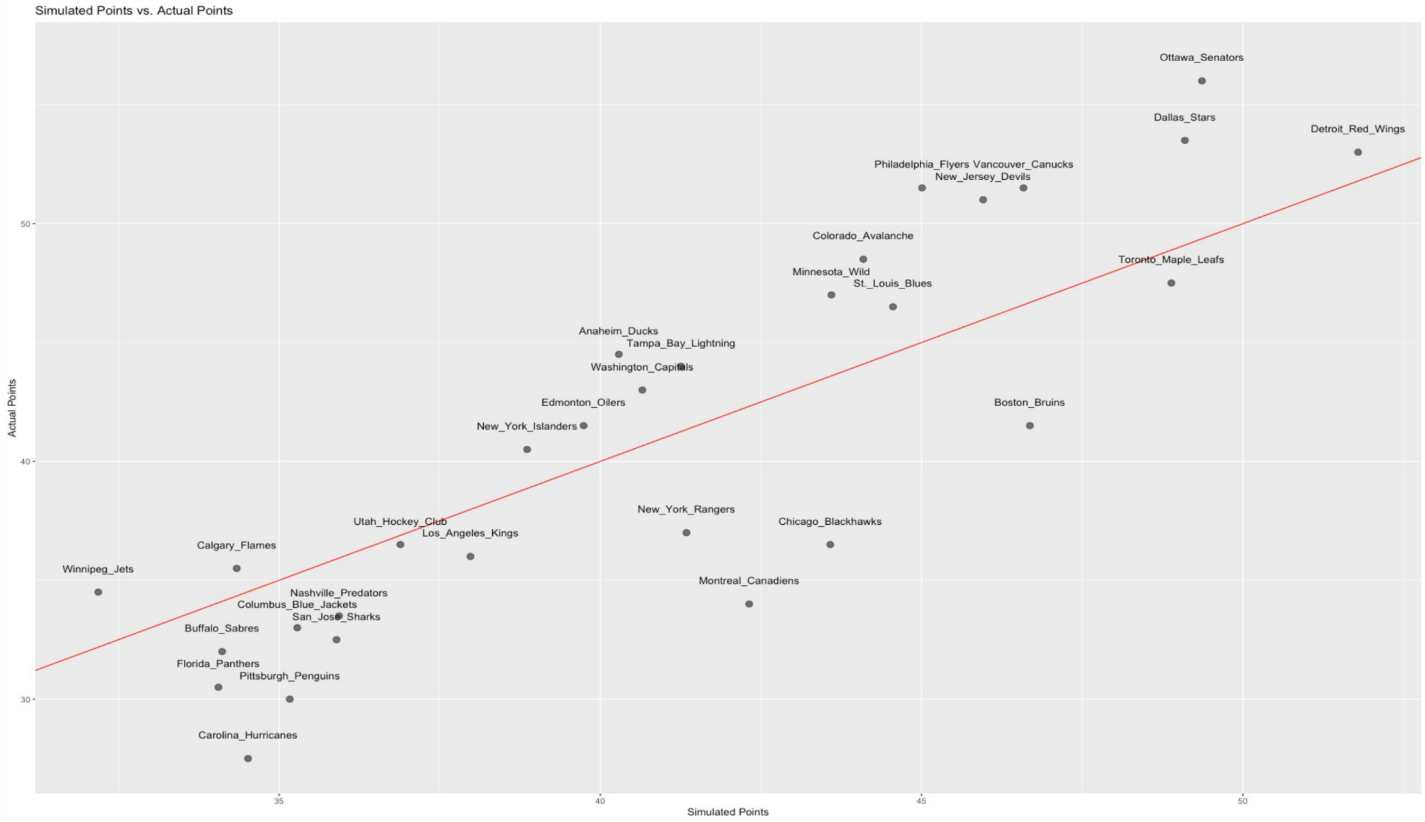
Team	Actual Wins	Simulated Wins
Dallas Stars	53.5	49.098
Detroit Red Wings	53	51.793
Edmonton Oilers	41.5	39.740
Florida Panthers	30.5	34.055
Los Angeles Kings	36	37.978
Minnesota Wild	47	43.596
Montreal Canadiens	34	42.315
Nashville Predators	33.5	35.933

Monte Carlo Simulation

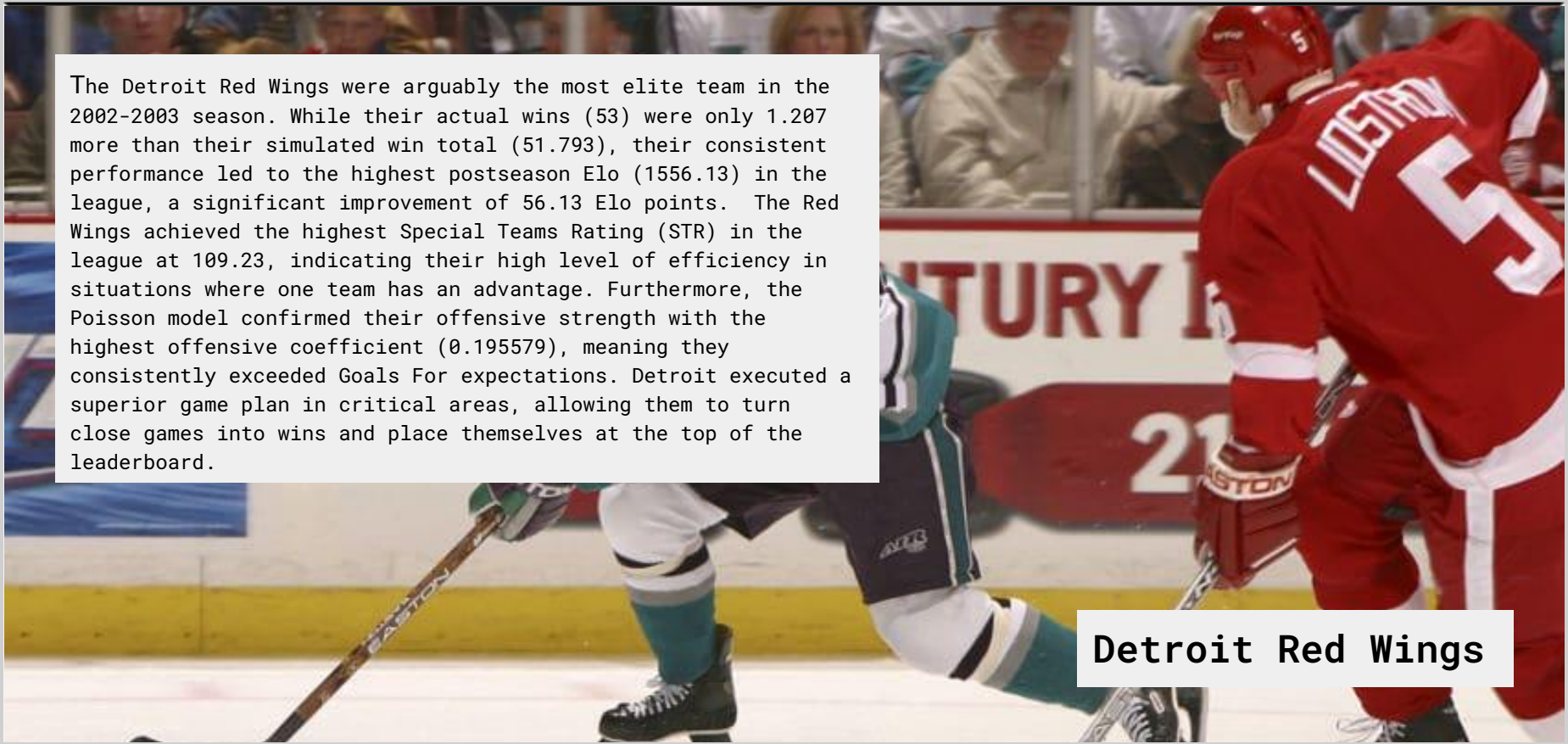
Team	Actual Wins	Simulated Wins
New Jersey Devils	51	45.957
New York Islanders	40.5	38.860
New York Rangers	37	41.340
Ottawa Senators	56	49.363
Philadelphia Flyers	51.5	45.007
Pittsburgh Penguins	30	35.166
San Jose Sharks	32.5	35.894
St. Louis Blues	46.5	44.554

Team	Actual Wins	Simulated Wins
Tampa Bay Lightning	44	41.257
Toronto Maple Leafs	47.5	48.887
Utah Hockey Club	36.5	36.888
Vancouver Canucks	51.5	46.585
Washington Capitals	43	40.653
Winnipeg Jets	34.5	32.185
Tampa Bay Lightning	44	41.257
Toronto Maple Leafs	47.5	48.887

Monte Carlo Simulation



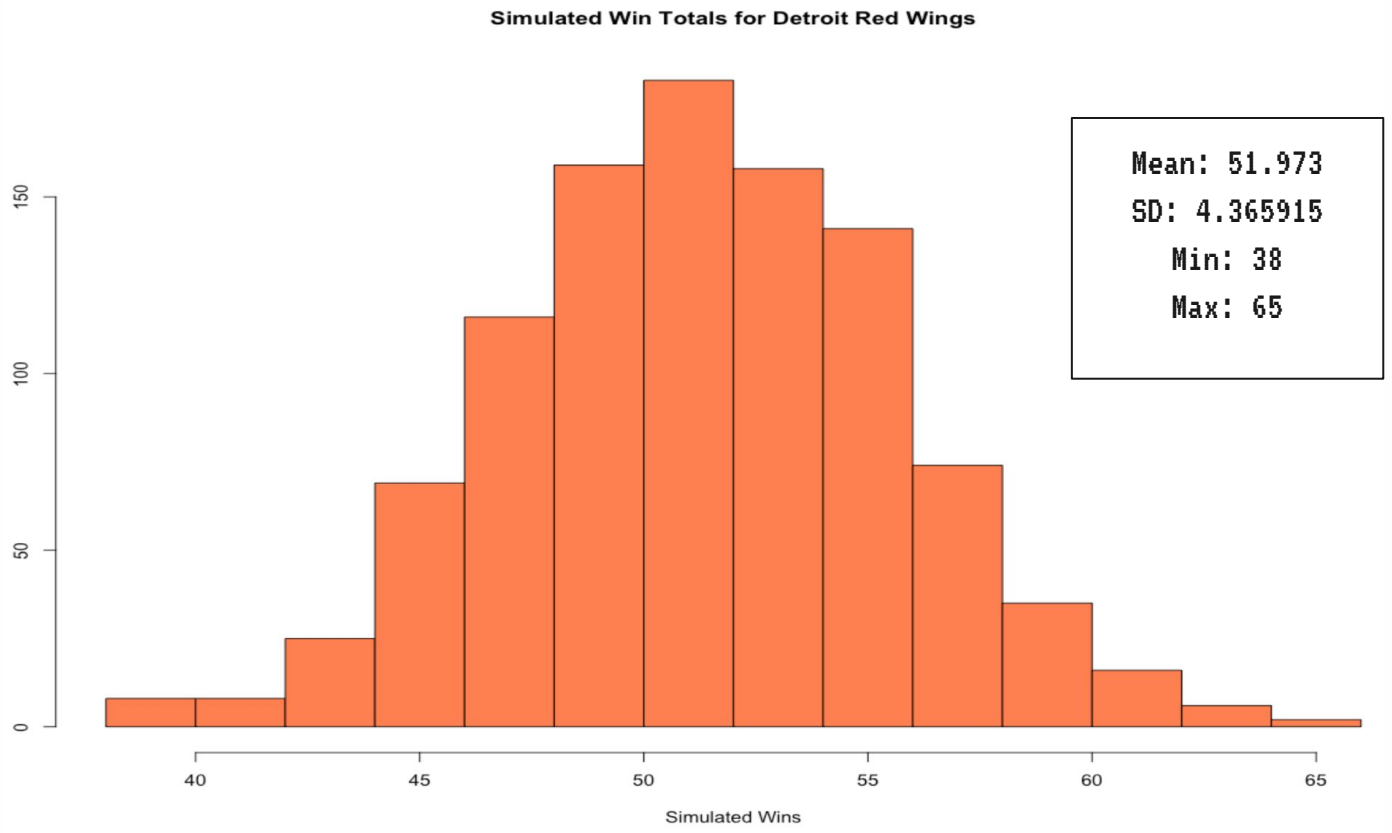
Analysis of Overachieving Team



The Detroit Red Wings were arguably the most elite team in the 2002-2003 season. While their actual wins (53) were only 1.207 more than their simulated win total (51.793), their consistent performance led to the highest postseason Elo (1556.13) in the league, a significant improvement of 56.13 Elo points. The Red Wings achieved the highest Special Teams Rating (STR) in the league at 109.23, indicating their high level of efficiency in situations where one team has an advantage. Furthermore, the Poisson model confirmed their offensive strength with the highest offensive coefficient (0.195579), meaning they consistently exceeded Goals For expectations. Detroit executed a superior game plan in critical areas, allowing them to turn close games into wins and place themselves at the top of the leaderboard.

Detroit Red Wings

Analysis of Overachieving Team

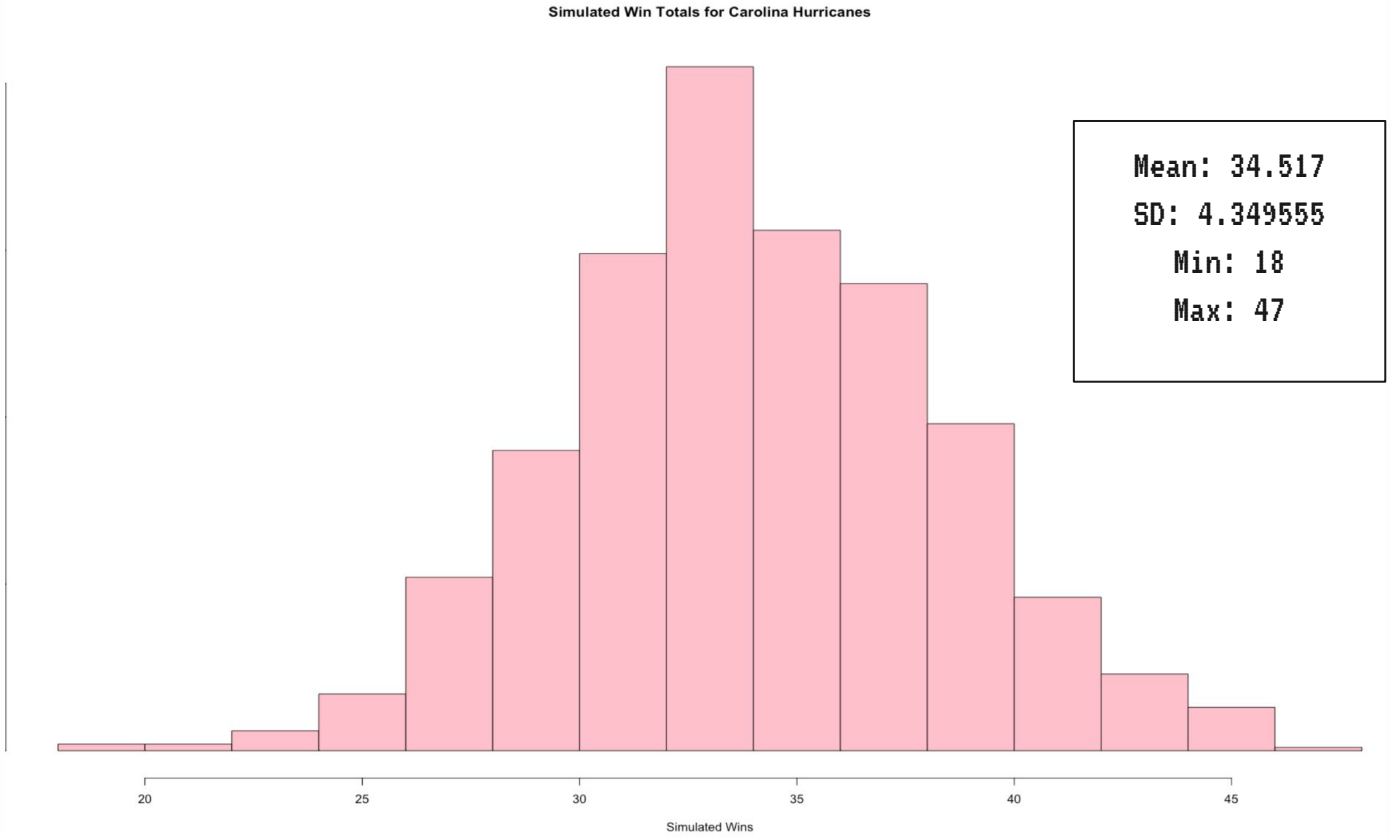


Analysis of Underachieving Team

Carolina Hurricanes

The Carolina Hurricanes represent the season's overall underachiever. While the Montreal Canadiens had the largest negative win residual, the Hurricanes' season was statistically a failure by every other metric. The Hurricanes had the largest Elo decline of -145.49 points, which culminated in the worst postseason Elo rating at 1254.507. Their 27.5 actual wins were 7.017 less than their simulated win total (34.517), potentially due to poor offense or bad luck. The Hurricanes had the most negative offensive coefficient (-0.270145) in the Poisson Scoring model and the worst PDO (96.8). In addition, they had one of the lowest special teams ratings (95.70), indicating they did not handle advantages efficiently. Overall, the Hurricanes were unable to succeed in critical areas. The combined weak offense, inefficiency, and bad luck explain why their final results were worse than the Elo and Monte Carlo simulations predicted, leading to their plummet to the bottom of the rankings.

Analysis of Underachieving Team



Key Takeaways

I

Goal differential was the dominant predictor of regular season success

II

Success and Failure were driven by measurable inefficiencies

III

The Anaheim Ducks' Cinderella run was a statistical anomaly

I. Goal Differential

The analytical models overwhelmingly confirmed that a team’s scoring margin was the most reliable indicator of its final record.

- The correlation between Pythagorean point percentage and actual point percentage was an extremely high 0.983, showing that goal differential perfectly explains nearly all variation in regular-season wins.
- The Monte Carlo simulation accurately predicted % division winners and the Elo model’s final rankings closely mirrored the seasons final standings.
- The top teams in the final postseason Elo ratings were also leaders in the Simple Rating System (SRS), validating their league dominance.

II. Inefficiencies

The analysis of over- and under-performing teams reveals long-term outcomes were largely dictated by specific skills captured by the STR and Poisson Offensive Coefficient

- The Detroit Red Wings STR (109.23) and highest offensive coefficient (0.195579) established their elite status in the league. Their success was built on efficiencies in high-leverage situations, not on luck.
- The Carolina Hurricanes exhibited the most negative offensive coefficient (-0.270145) and lowest STR (95.70), which was the primary driver of their 145.49 Elo drop.
- The Hurricanes failure was compounded by having the season’s worst PDO (96.8), validating that their bad luck combined with inefficiencies led to a disaster.

II. Anaheim Ducks

The final outcome of the Stanley Cup playoffs contrasted with the regular-season established hierarchy.

- Despite the Elo and simulation models identifying clear “elite” groups, the 7th-seeded Anaheim Ducks made it to the Stanley Cup final with extreme Elo variations.
- The Ducks overachieved their Monte Carlo simulation expectations, winning 4.213 more games than the simulated 40.287.
- The Ducks had a below-average offense (-0.0943), but one of the best defenses (0.4386) in the league, crucial for high-leverage situations
- The run was largely credited to Jean-Sébastien Giguère, the Ducks’ exceptional goaltender.

Thank you !

References:

Stats obtained from
https://www.hockey-reference.com/

nhl_data.csv
nhl_initial_elos.csv

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